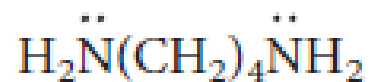


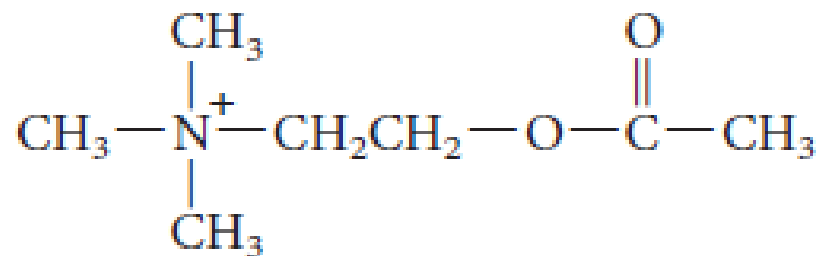
Amines and Related Nitrogen compounds



putrescine

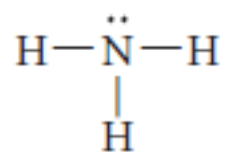


1,6-diaminohexane

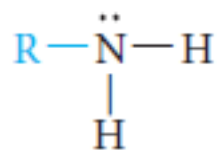


acetylcholine

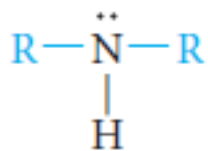
11.1 Classification and Structure of Amines



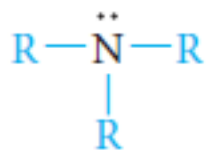
ammonia



primary amine

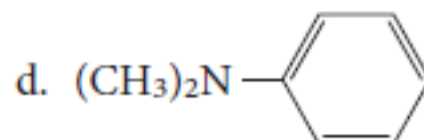
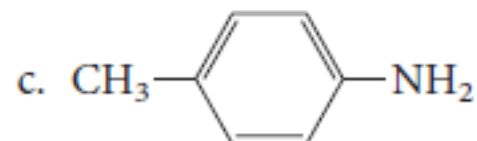
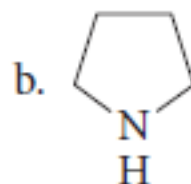
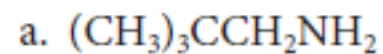


secondary amine

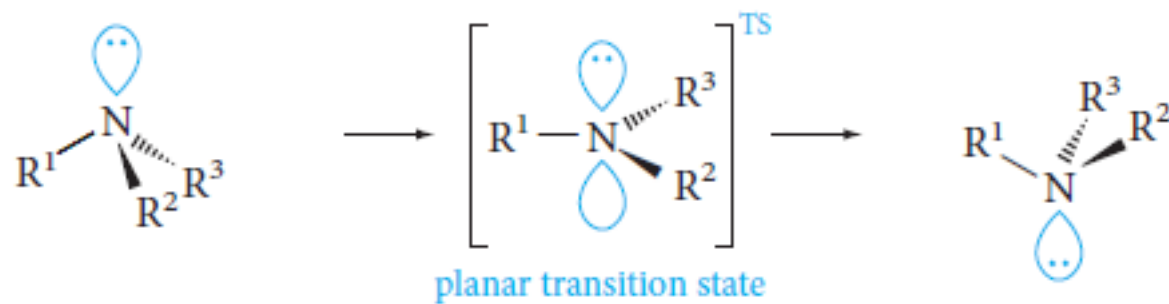
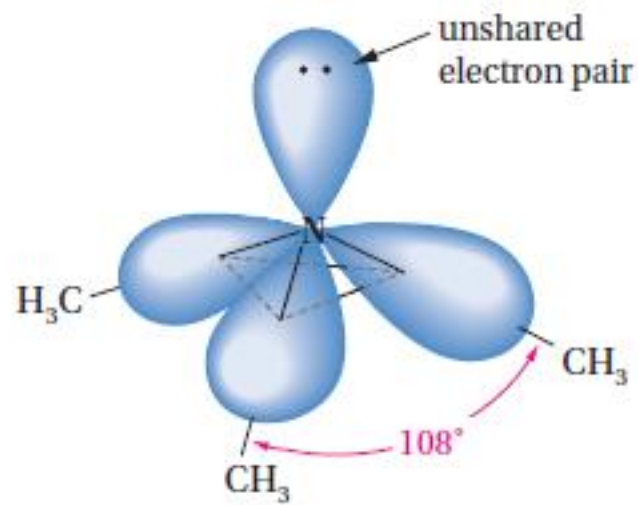


tertiary amine

R groups in these structures may be alkyl or aryl



The nitrogen orbitals are sp^3 hybridized



11.3 Physical Properties and Intermolecular Interactions of Amines

Intermolecular $\text{N—H} \cdots \text{N}$ hydrogen bonds are important and raise the boiling points of primary and secondary amines but are not as strong as the $\text{O—H} \cdots \text{O}$ bonds of alcohols

Table 11.1  The Boiling Points of Some Simple Amines

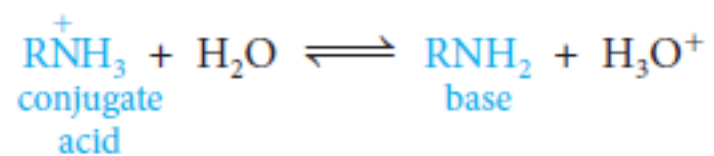
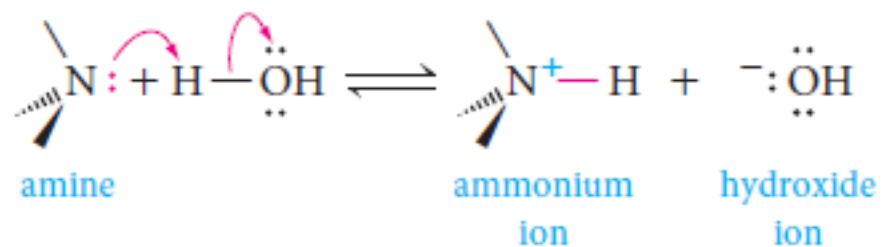
Name	Formula	bp, °C
ammonia	NH_3	−33.4
methylamine	CH_3NH_2	−6.3
dimethylamine	$(\text{CH}_3)_2\text{NH}$	7.4
trimethylamine	$(\text{CH}_3)_3\text{N}$	2.9
ethylamine	$\text{CH}_3\text{CH}_2\text{NH}_2$	16.6
propylamine	$\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$	48.7
butylamine	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$	77.8
aniline	$\text{C}_6\text{H}_5\text{NH}_2$	184.0

Table 11.2  **A Comparison of Alkane, Amine, and Alcohol Boiling Points***

alkane	CH_3CH_3 (30) bp -88.6°C	$\text{CH}_3\text{CH}_2\text{CH}_3$ (44) bp -42.1°C
amine	CH_3NH_2 (31) bp -6.3°C	$\text{CH}_3\text{CH}_2\text{NH}_2$ (45) bp $+16.6^\circ\text{C}$
alcohol	CH_3OH (32) bp $+65.0^\circ\text{C}$	$\text{CH}_3\text{CH}_2\text{OH}$ (46) bp $+78.5^\circ\text{C}$

*Molecular weights (in g/mol) are given in parentheses.

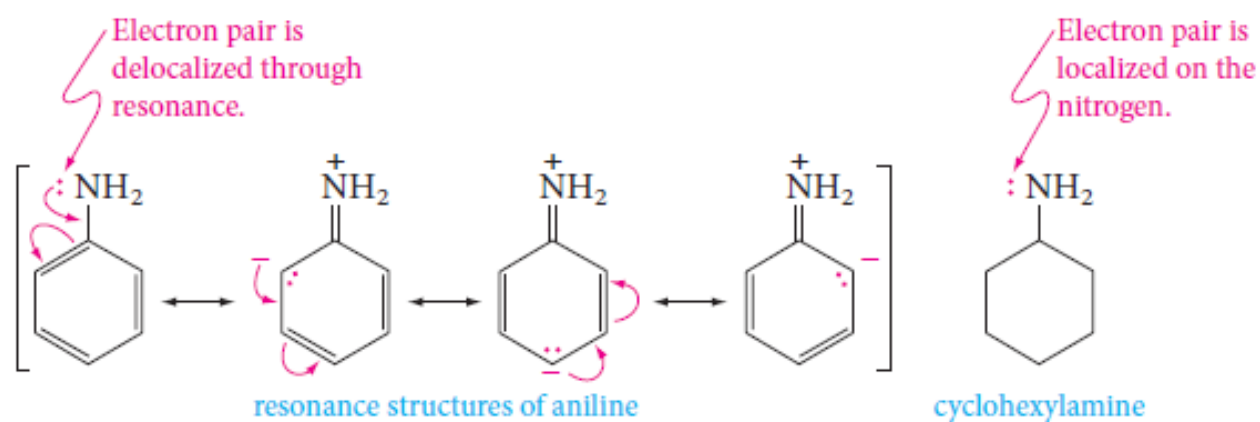
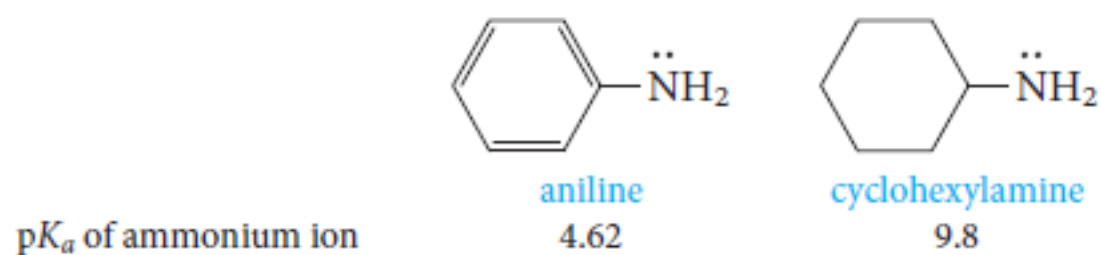
11.6 The Basicity of Amines



$$K_a = \frac{[\text{RNH}_2][\text{H}_3\text{O}^+]}{[\text{RNH}_3^+]}$$

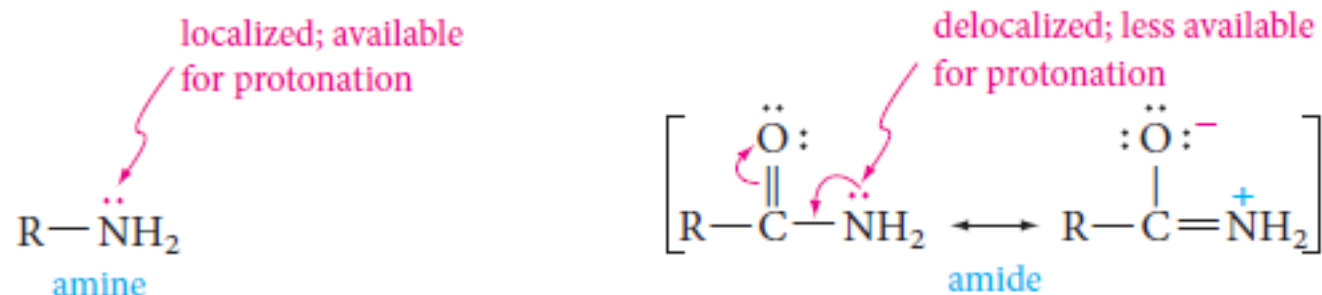
In general, *electron-donating groups increase the basicity of amines, and electron-withdrawing groups decrease their basicity.*

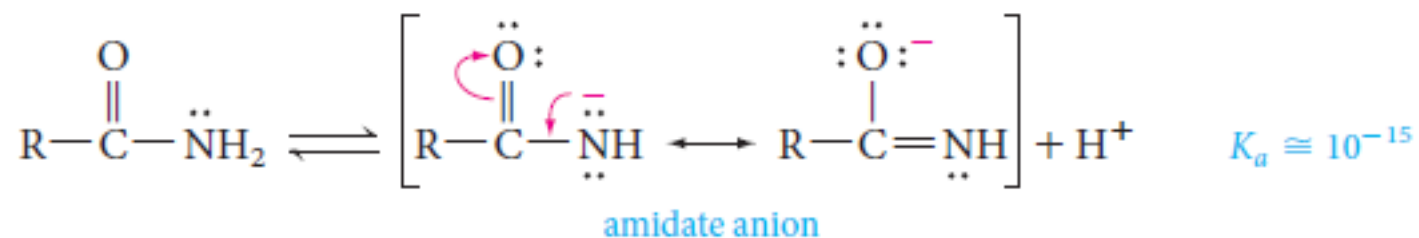
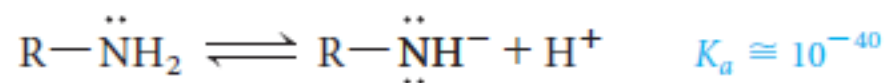
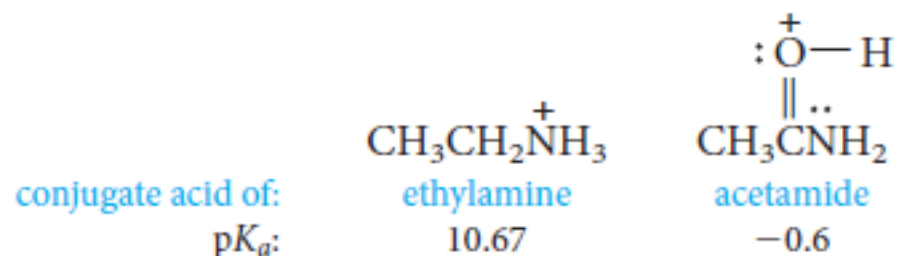
Aromatic amines are much weaker bases than aliphatic amines or ammonia. For example, aniline is less basic than cyclohexylamine by nearly a million times.



11.7 Comparison of the Basicity and Acidity of Amines and Amides

Amines and amides each have nitrogens with an unshared electron pair. There is a huge difference, however, in their basicities. Aqueous solutions of *amines* are *basic*; aqueous solutions of *amides* are essentially *neutral*. Why this striking difference?





PROBLEM 11.16 Place the following compounds (a) in order of increasing basicity and (b) in order of increasing acidity.

